



Composition of Asteroids and Their Formation and Evolution Within the Solar System



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Abstract

The mineralogical composition of asteroids can indicate a further understanding of the history and evolution of the solar system. While prior research has been conducted on the compositions of specific asteroids, this research analyzes multiple classifications of asteroids to create a broader narrative on the formation and arrangement of the solar system. This research focuses on five different types of asteroids along with an example of each particular type: C-type Ceres, M-type Psyche, B-type Benu, S-type Itokawa, and V-type Vesta. Focusing on the mineralogy, age, density, and structure in comparison to each asteroid's relative location in the solar system and to the Sun, the research concluded that the mineralogical composition of asteroids is highly correlated to their location within the solar system. The distance from the sun and the resulting thermal energy play a large role in the differentiation of the minerals and their location within the structure of the asteroid. Furthermore, analysis of these asteroids revealed important links between their compositions and their origins. The mineralogy of the asteroids studied provided detailed insight into where they originated and the conditions of where they formed. In addition to the upcoming Psyche sample results and OSIRIS-APEX mission, possible future research in this area could include analysis of Kuiper Belt objects in a similar manner to that which is done in this research. This would help further scientific understanding of the formation of objects on the fringes of our solar system by drawing comparisons to those already studied.

Discussion

Vesta

- ★ Formed via dust accretion in early protoplanetary disk
- ★ HED (howardites, eucrites, & diogenites) asteroids originate from Vesta
- ★ We know this because of similarities in compositions and spectra
- ★ Small differences exist between subtypes due to core differentiation and ejection times

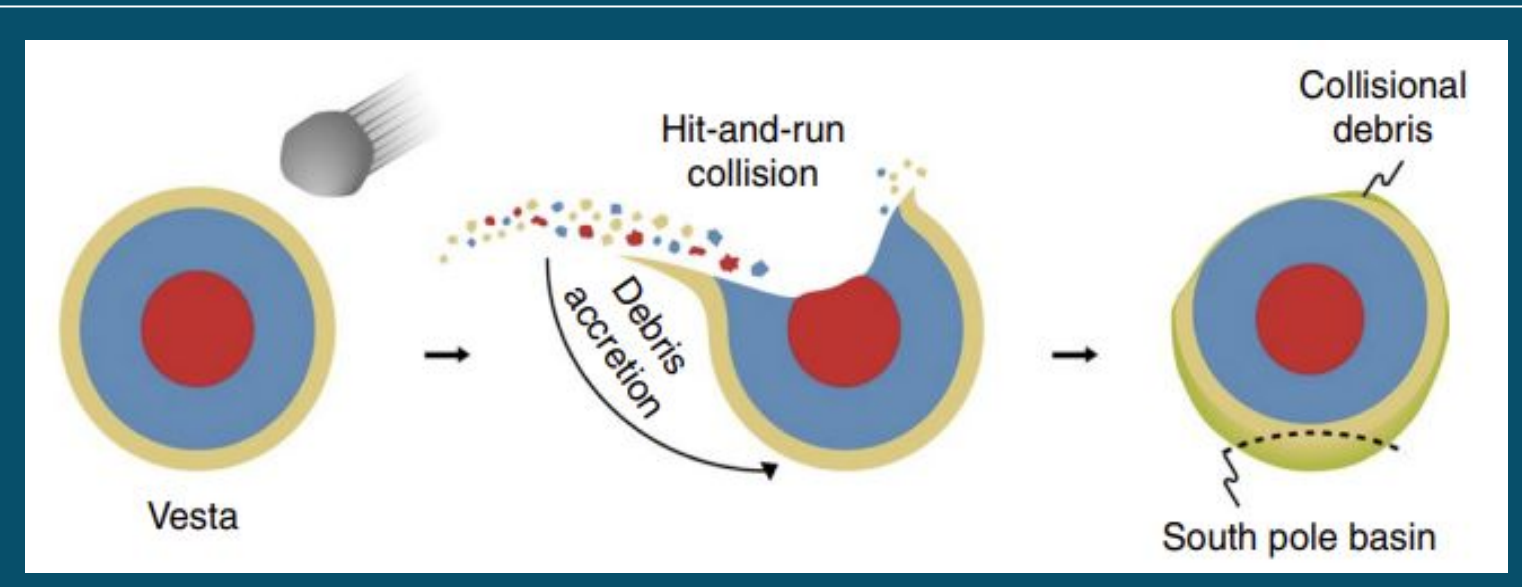


Fig. 4: The ejection of the HED group of asteroids (4)

Psyche

- ★ Composed mostly of nickel-copper alloy, silicates, and iron
- ★ Started out fully molten, solidified over time
- ★ It has been impacted various times, making its surface rough and porous

Ceres

- ★ Carbon composition → dark charcoal colour
- ★ Comprised of silicate substances accumulated on celestial bodies with surface water near Jupiter and Saturn
 - Partially made up of silicate rocks, silicate organic matter, clay
 - 22% made up of water

Benu

- ★ OSIRIS-REx samples show evidence of organic molecules
- ★ Impacts from asteroids similar in composition → the source of organic molecules on Earth / life on our planet

Itokawa

- ★ Low density on account of formation
- ★ Low mass □ Low Gravity □ Agglomeration
- ★ Parent Body
 - C-Type with Water + Organic compounds
- ★ Itokawa ~ S-Type
 - Mostly silicates



Fig. 5: Large 50m boulder on Itokawa that could not have formed from any impact to Itokawa (5)

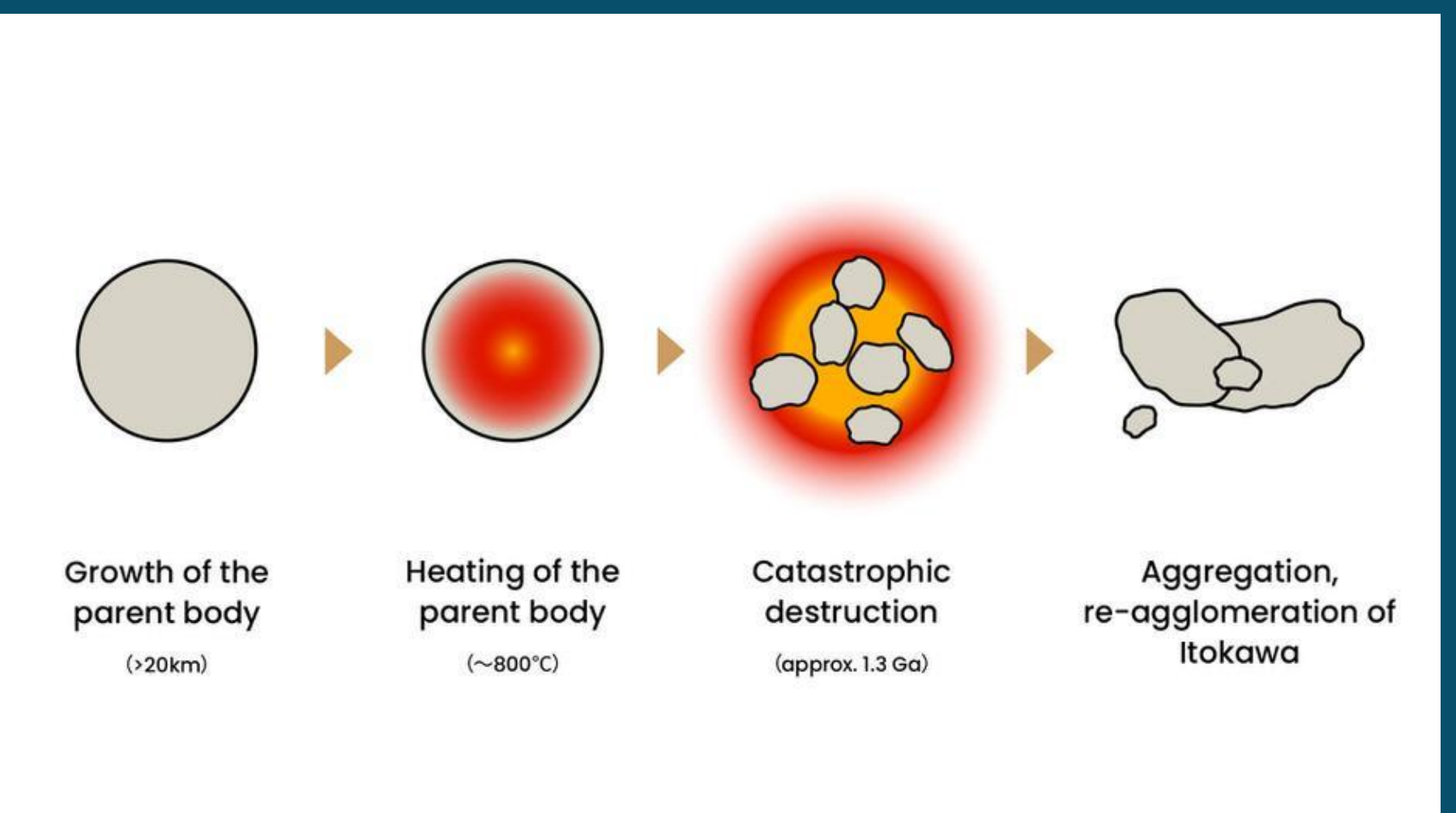


Fig. 6: The formation of Itokawa including the parent body, metamorphosis, impact, and re-agglomeration (6)

Introduction

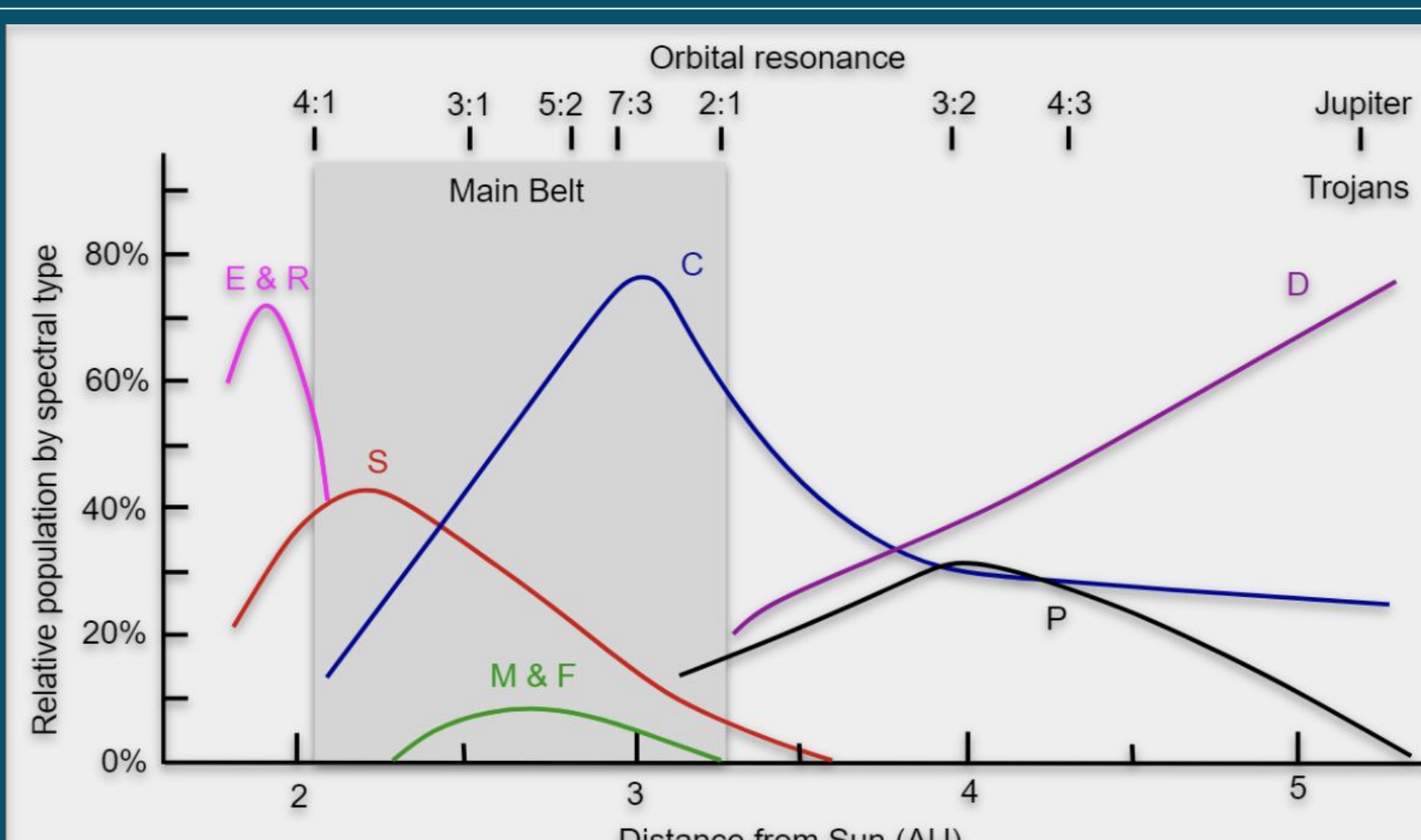


Fig. 1: A graphical analysis comparing asteroid types and their relative populations (1)

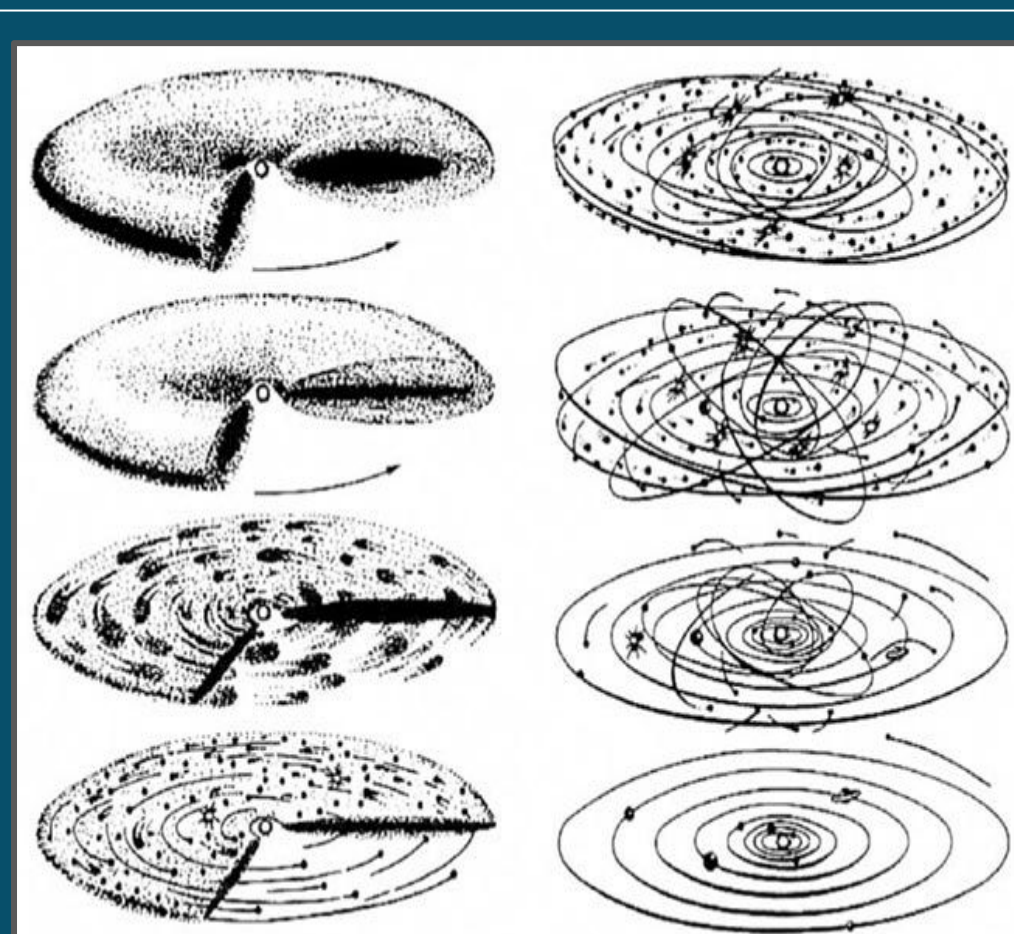


Fig. 2: Particles in stellar nebula spin and gravitate together, initiating nuclear fusion to create stars and colliding to create planets (2)

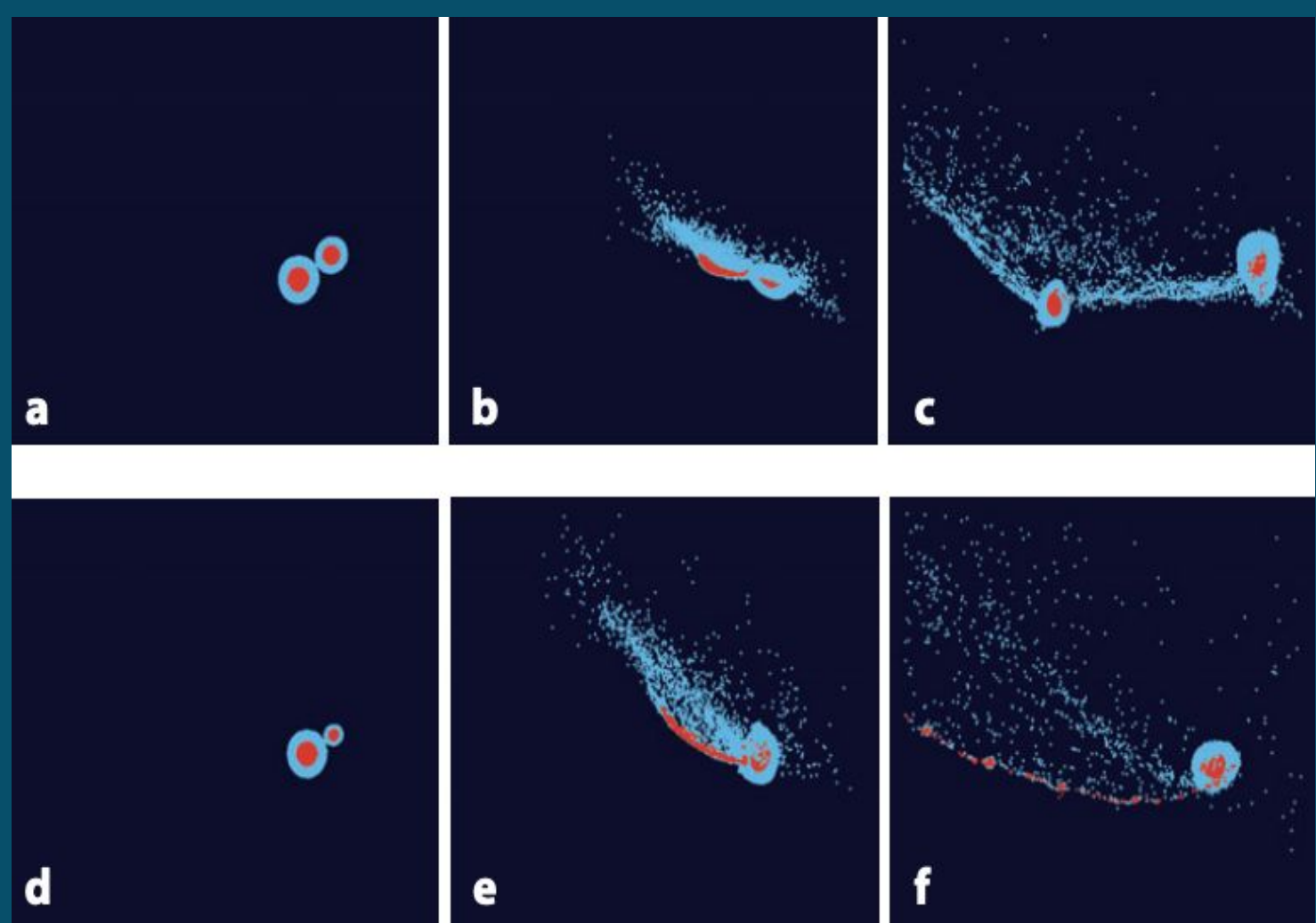


Fig. 3: A diagram of two instances (top and bottom row) where two asteroids collide into each other, forming different objects/asteroids and scattering debris into the solar system (3)

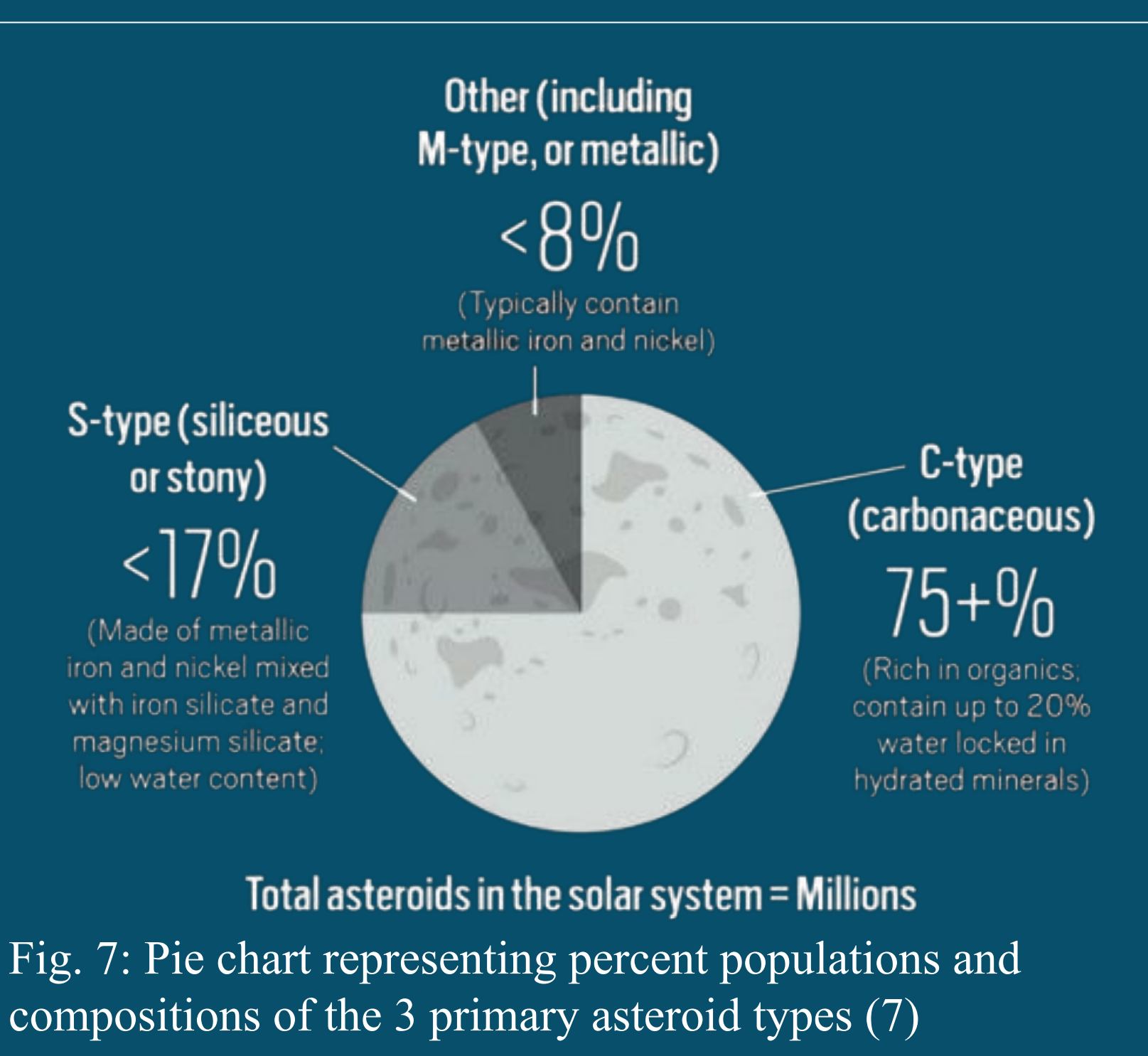
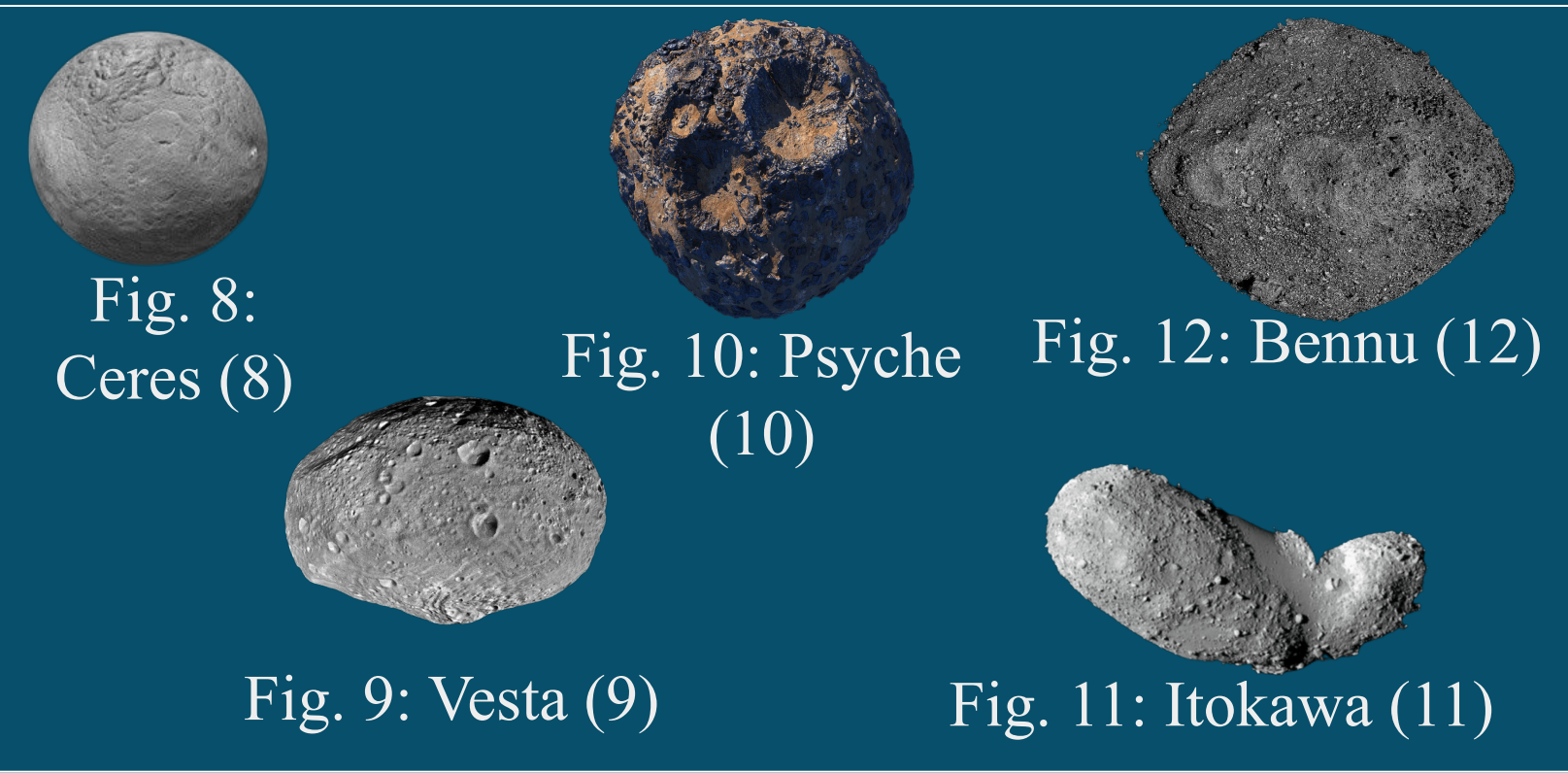


Fig. 7: Pie chart representing percent populations and compositions of the 3 primary asteroid types (7)



Methodology

- ★ Scholarly resources - Google scholar, EBSCO, NASA, JPL, various universities and accredited institutions
- ★ Chose each asteroid due to their unique characteristics as well as the amount of existing research

Conclusion

This research provides:

- ★ Composition helps determine relative locations of asteroids
- ★ Further understanding of development of solar system
- ★ Hypothesis about development of life on Earth
- ★ Less resilient carbonaceous chondrite composites

Future Work

The Psyche probe will study asteroid Psyche with satellite. As the first mission to study a metallic asteroid in 2029, it will expand information on M-type asteroids

OSIRIS-APEX will study the asteroid Apophis observing changes in activity and its composition providing valuable data to be analyzed

The Kuiper Belt, as an analog to the asteroid belt, could be studied for analysis of its location, structures, and effect on the formation of the solar system

Results

	Ceres	Vesta	Psyche	Benu	Itokawa
Composition	Water-ice mantle, sodium carbonate, ammonium chloride, organic compounds and other rocks with the composition of C1 chondrites (13)	Basalt, nickel-iron, silicates (olivine and plagioclase), pyroxenes (orthopyroxene and clinopyroxene), and oxide minerals (ilmenite and chromite) (14)	Nickel-copper (Monel), silicate, iron (15)	Serpentine, Magnesium sodium Phosphate, organic compounds and clay minerals (16)	Silicates, water and organic compounds (17)
Density	2700 kg/m ³ (13)	3456 kg/m ³ (14)	2950±1550 kg/m ³ (15)	1260 ± 70 kg/m ³ (16)	1900 ± 130 kg/m ³ (17)

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